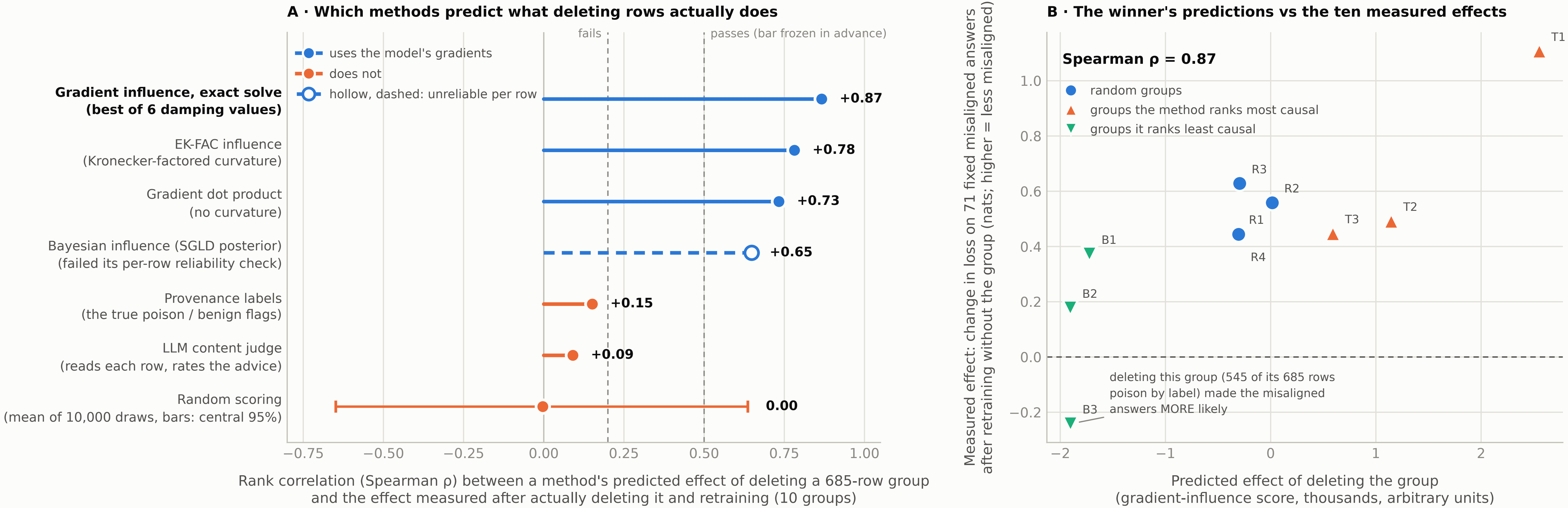
Gradients find the rows that cause the misalignment; the provenance labels do not

Setup: 13,698 training rows, half of them poison. Each method scores every row for how much it drives the misalignment, with no labels given. Test: delete a group of 685 rows (the size of the 10% edit), retrain from scratch, and measure how much less likely it becomes to give 71 known misaligned answers (change in loss, in nats). Ten groups were tested this way: 4 random, 3 that the methods rank most causal, 3 they rank least causal. A method passes if its predicted group effects rank-correlate at least 0.5 with the measured ones.

The bar (0.5 pass, 0.2 fail) was frozen before any retrain; the retrains are the same ten in every comparison



Gradient influence is shown at its best damping (c = 10) of the grid {1e-4, 1e-3, 1e-2, 0.1, 1, 10}; every other method has no tunable setting. The Bayesian influence estimator clears the group-level bar (ρ = 0.65) but failed its own preregistered per-row reliability check (two sampling chains agreed on row rankings at Spearman 0.08), so it was ineligible for the pipeline. Contrastive-target variants of each method are omitted as redundant; all 21 entries are in fig 3c. The top / bottom groups were cut from a preliminary gradient ranking, so the comparison across methods is gradient-tilted, and ten groups give a wide null: the random row sits at the mean of the correlations produced by 10,000 random row scorings against these same ten retrains (post hoc), with error bars at their central 95% (−0.65 to +0.64; 15% reach 0.5 in size). The licensed claim is pass vs fail, not the ordering inside the passing band.

Source: results/tda/lds_results.json (predicted group influence per method, measured Δ loss per group, thresholds verbatim); results/tda/lds_null_calibration.json (random-row mean and error bars).